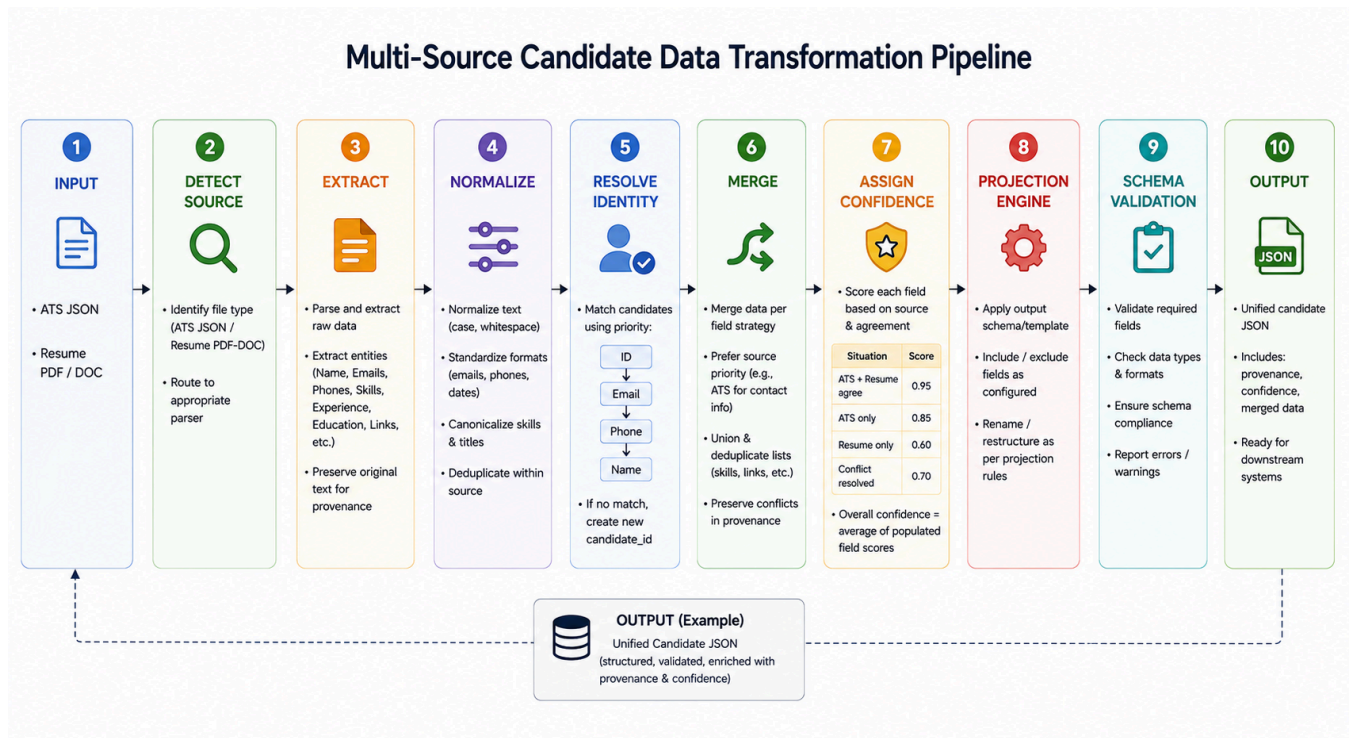


Technical Design - CandidateFusion - Multi Source Candidate Data Transformer

1. Pipeline



Design Principles

- Pure deterministic pipeline; each stage is independently testable.
- Canonical record is immutable after merge.
- Projection layer only reshapes output.
- Missing or corrupted text sources never stop execution.

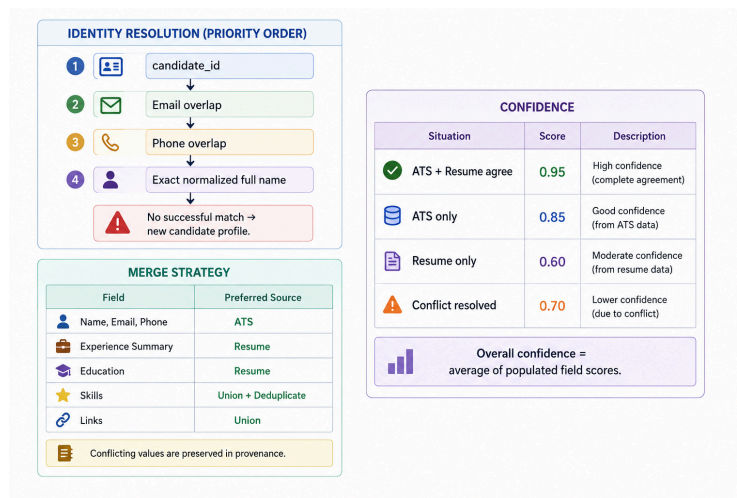
2. Canonical Candidate Schema

```
Candidate {  
  candidate_id, full_name,  
  emails[], phones[],  
  location { city, region, country },  
  links { linkedin, github, portfolio, other[] },  
  headline, years_experience,  
  skills[] { name, confidence, sources[] },  
  experience[], education[],  
  provenance[],  
  overall_confidence  
}
```

Normalization

- Emails → lowercase + deduplicated
- Phones → E.164
- Dates → YYYY-MM (unknown month never invented)
- Countries → ISO-3166 Alpha-2
- Skills → canonical synonym mapping
- Unknown values remain **null**, never fabricated

3. Identity Resolution, Merge & Confidence



4. Runtime Projection Engine

Single configurable engine (default output is simply the default configuration).

Supports:

- Field selection
- Field renaming (**from**)
- Runtime normalization
- Include/omit confidence & provenance
- Missing-value policy (**null**, **omit**, **error**)
- JSON Schema validation

5. Robustness & Edge Cases

- Unknown ATS fields → ignored and logged
- Corrupted/invalid resumes → processing continues
- Missing values → null (never guessed)
- Empty arrays remain empty
- Duplicate emails/phones/skills removed
- Invalid dates or locations lower confidence but never stop execution

6. Constraints Satisfied

- Deterministic & Explainable: Same input always produces identical output, with every value carrying its source and transformation history.
- Robust: Missing or noisy inputs never crash the pipeline; unknown information is preserved as null and never fabricated.
- Scalable: Linear-time processing per candidate with independent stages enabling parallel ingestion.

Design Philosophy

Maintain a single immutable canonical candidate profile while separating ingestion, normalization, conflict resolution, and projection. Every output is deterministic, explainable, traceable to its origin, and configurable at runtime without changing the transformation engine.